COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 10 : SLO – 1 | NOR gate and universal implementation
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Part B – Multiple Choice Questions

• 1. B — (A + B)'
• 2. C — It can implement AND, OR, NOT gates
• 3. B — Connect inputs together
• 4. B — (A' + B')'
• 5. D — All of the above

Part A – Match the Following

• 1. NOR gate — c) Output is 1 only when all inputs are 0
• 2. NOR as NOT — a) Connect both inputs together
• 3. NOR as OR — NOR the inputs, then invert the output with another NOR: ((A+B)')' = A + B
• 4. NOR as AND — e) De Morgan: invert each input first, (A' + B')' = A·B
• 5. Universal Gate — d) Any gate can be built using it

Part C – Activity Questions

• 1. 2-input NOR truth table: 00→1, 01→0, 10→0, 11→0.
• 2. NOR as NOT: tie both inputs together → Y = (A+A)' = A■. Truth table: 0→1, 1→0.
• 3. AND using only NOR gates: invert A (NOR with tied inputs) → A'; invert B → B'; then NOR the two results: (A' + B')' = A·B. Total 3 NOR gates.